Automated UI Testing for an E-Commerce Website

1. Literature Review

In the rapidly evolving landscape of e-commerce, ensuring a seamless user experience is critical for customer retention and maximizing revenue. Automated UI testing plays a vital role in verifying that the visual and interactive elements of a web application function correctly across different browsers and platforms.

Several automated testing frameworks exist in the industry, including Cypress, Playwright, and Selenium. For this project, **Selenium WebDriver** was selected as the primary testing tool. Selenium remains an industry standard due to its robust cross-browser compatibility, extensive community support, and flexibility in integrating with multiple programming languages such as Python and Java. Unlike some modern tools that are limited to specific environments, Selenium's architectural design allows for comprehensive end-to-end testing of complex e-commerce workflows.

Furthermore, maintaining large test suites can become challenging as the application scales. To address this, the project adopts the **Page Object Model (POM)** design pattern. POM is widely recognized in software testing literature as a best practice. It separates the UI element locators (the web page representation) from the actual test scripts. This separation of concerns significantly enhances code reusability, reduces duplication, and makes the automation framework highly maintainable when UI changes occur.